David S. Medina

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EDUCATION

Graduate Student in Computational and Applied Mathematics, Rice University — June 2011 - May 2015

- *Advisor:* Prof. Tim Warburton
- *Ph.D Thesis:* OKL*: A unified kernel language for parallel architectures* — May 2015
- *Master's Thesis:* OCCA*: A unified approach to multi-threading languages* — April 2014

Bachelor of Science in Mathematics, University of Texas Pan-American, TX — August 2007 - May 2011

- Double major in Mathematics and Computer Science

RESEARCH INTERESTS

- Parallel programming implementations of numerical methods (FD/FEM/DG)
- Computational graphics

PUBLICATIONS

- *Portability and Performance of Nuclear Reactor Simulations on Many-Core Architectures* (Submitted) May 2015

  Rahaman, R., **Medina, D.**, Lund, A., Tramm, J., Warburton, T. and Seigel, A. *77th EAGE Conference and Exhibition 2015*

- *A Model Evaluation Study for Treatment Planning of Laser Induced Thermal Therapy* (Submitted) May 2015

  Fahrenholtz, S. J., Moon, T., Franco, M., **Medina, D.**, Danish, S., Gowda, A., Shetty, A., Maier, F., Hazle, J., Stafford, R. J., Warburton, T. and Fuentes, D.

- *GPU Accelerated discontinuous Galerkin methods for shallow water equations* — May 2015

  Gandham, R., **Medina, D.** and Warburton, T., *Communications in Computational Physics*

  ArXiv preprint can be found at http://arxiv.org/abs/1403.1661

- *High-Order Finite-differences on multi-threaded architectures using* OCCA — October 2014

  Medina, D. S., St-Cyr, A. and Warburton, T., ICOSAHOM 2014

  ArXiv preprint can be found at http://arxiv.org/abs/1410.1387

- OCCA*: A unified approach to multi-threading languages* — February 2014

  Medina, D. S., St-Cyr, A. and Warburton, T.

  ArXiv preprint can be found at http://arxiv.org/abs/1403.0968

POSTERS AND CONFERENCE TALKS

- Talk on OKL*: A unified kernel language for parallel architectures* — March 2015

  Given at Sandia National Laboratory

- Poster and Talk on OKL*: A unified kernel language for parallel architectures* — March 2015

  Given at *SIAM CSE '15*

- Tutorial and Talk on OKL*: A unified kernel language for parallel architectures* — March 2015

Given at *Rice Oil & Gas '15* (Rice University)

- Talk on *Tutorial on* OCCA — December 2014

  Given at a CESAR Sound-off meeting (Argonne National Lab)

- Talk on OCCA*: A unified approach to multi-threading languages* — September 2014

  Given at Computation-Institute for Scientific Computing Research, Lawrence Livermore National Lab

- Talk on OCCA*: A unified approach to multi-threading languages* — June 2014

  Presented at "ICOSAHOM 2014"

- Poster on OCCA*: A unified approach to multi-threading languages* — March 2014

  David S Medina, Amik St-Cyr, T. Warburton presented at "Oil & Gas HPC Workshop" at Rice University

- Poster on *GPU Accelerated Numerical Methods for Tsunami Modeling* — December 2013

  R. Gandham, D. Medina, T. Warburton presented at IEEE conference on "High Performance Computing" in Bengaluru

- *High-order Numerical Methods for High-Contrast Seismic Imaging* — February 2013

  R. Gandham, D. Medina, T. Warburton, Amik St-Cyr presented at "Oil & Gas HPC Workshop" at Rice University

PROJECTS

- OCCA[1]: Developed back-ends for OpenMP, Pthreads, OpenCL, CUDA and COI with native front-ends for C, C++, Python, Julia and Matlab. Work involved knowing low-level threading, memory hierarchies, managing JIT function compilations, and knowledge of accelerated hardware such as GPUs and Intel's Xeon Phi.
- RiDG: MPI + OCCA project involving discontinuous Galerkin for high-order seismic imaging
- gNek: MPI + OCCA project using spectral element method for high-order CFD simulations

TECHNICAL SKILLS

- Programming Languages and Standards: C, C++, OpenGL, GLSL
- Parallel Programming Languages and Standards: OpenCL, CUDA, MPI, OpenMP, COI
- Scripting Tools: Bash (shell), Python (general-purpose)

RESEARCH EXPERIENCE

Shell Summer Internship — Summer 2014

Shell Technology Center, Houston, TX

- Optimized finite difference codes on multiple architectures
- Investigated hybrid kernels with portable performance across varying architectures

Shell Contractor — Spring 2014 - Fall 2014

Shell Technology Center, Houston, TX

- Working on numerical methods in parallel architectures (OpenMP, OpenCL, and CUDA)

Shell Summer Internship — Summer 2013

Shell Technology Center, Houston, TX

[1] www.libocca.org

- Worked on Discontinuous Galerkin implementations of wave equations
- Applied a portability layer between different threading models (OpenCL, CUDA, C)

PASI: Workshop on Tsunami Modeling (NSF/DOE-Funded Travel Grant) January 2013
Valparaiso, Chile

- Attended multiple workshops and talks about tsunami modeling
- Lab co-coordinator to teach OpenCL/OpenGL interop for applying and visualizing finite difference

Staff in HPC Summer Institute Summer 2012
Rice University, Houston, TX

- Helped attendees with GPGPU programming

AWARDS

Ken Kennedy Institution of Technology Fall 2013 - Spring 2014
Rice University, Houston TX

Shell Fellowship Through the Ken Kennedy Institution of Technology Fall 2012 - Spring 2013
Rice University, Houston TX

LSAMP Award Fall 2010 - Spring 2011
University of Texas Pan-American, Edinburg, TX

University Excellence Scholarship Fall 2008 - Spring 2011
University of Texas Pan-American, Edinburg, TX